

Math 54 ReviewName: _____ ID: _____

Part I: Linear Systems (Chapter 1)**1. Systems and Row Reduction (1.1–1.2)****1.1** Solve the system

$$\begin{cases} x + 2y - z = 1, \\ 2x + 5y - 3z = 4, \\ x + 3y - 2z = 2. \end{cases}$$

(a) Write the augmented matrix. (b) Row reduce. (c) Give the solution set (or state none).

1.2 Determine whether the system is consistent. If consistent, describe *all* solutions.

$$\begin{cases} x - y + z = 0, \\ 2x - 2y + 2z = 3. \end{cases}$$

1.3 Row reduce the augmented matrix to RREF and solve:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 2 & 4 & 7 & 3 \\ 1 & 2 & 4 & 2 \end{array} \right].$$

Identify pivot columns and free variables.

1.4 For which values of k is the system consistent? For consistent k , describe the solution set.

$$\begin{cases} x + y + z = 1, \\ 2x + 2y + 2z = k. \end{cases}$$

2. Vector Equations and $Ax = b$ (1.3–1.5)**1.5** Decide whether

$$b = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} \quad \text{is in} \quad \text{span} \left\{ v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

If yes, find scalars c_1, c_2 such that $c_1v_1 + c_2v_2 = b$.

1.6 Let

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 3 \end{bmatrix}.$$

- (a) Compute Ax for $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and rewrite Ax as a linear combination of the columns of A .
- (b) Describe all $b \in \mathbb{R}^2$ for which $Ax = b$ is consistent.
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1.7 Solve the homogeneous system and give the solution in parametric vector form:

$$\begin{cases} x + 2y - z = 0, \\ 2x + 4y - 2z = 0. \end{cases}$$

1.8 (Structure of solutions) Consider

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}.$$

- (a) Find one particular solution x_p to $Ax = b$.
- (b) Find the general solution to $Ax = b$ in the form $x = x_p + x_h$, where $x_h \in \text{Null}(A)$.
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3. Linear Independence and Linear Transformations (1.7–1.9)

1.9 Determine whether the vectors are linearly independent:

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

If dependent, exhibit a nontrivial linear relation.

1.10 Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $T(x, y) = (2x - y, x + 3y)$.

- (a) Find the standard matrix A such that $T(\mathbf{x}) = A\mathbf{x}$.
- (b) Compute $T(1, 2)$.
- (c) Determine whether T is one-to-one and/or onto.
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1.11 Find the standard matrix of $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$T(x, y) = (x + 2y, 3x - y).$$

Then compute $T(2, -1)$.

Part II: Matrix Algebra (Chapter 2: 2.1–2.3)

2.1 Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}.$$

Compute $A + B$, $3A$, AB , and BA . Are AB and BA equal?

2.2 Compute A^2 and A^3 for

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

Conjecture a pattern for A^{2n} and A^{2n+1} .

2.3 Find A^{-1} (if it exists) using row reduction:

$$A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}.$$

Then solve $Ax = b$ for $b = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ using A^{-1} .

2.4 (Concept) True or False. Justify briefly.

- (a) If the columns of A are linearly independent, then A is invertible.
 - (b) If A is invertible, then $Ax = b$ has at most one solution for every b .
 - (c) If $Ax = 0$ has only the trivial solution, then A is invertible.
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Part III: Determinants (Chapter 3: 3.1–3.3)

3.1 Compute

$$\det \begin{bmatrix} 1 & 2 & 0 \\ 3 & 4 & 5 \\ 0 & 1 & 2 \end{bmatrix}$$

using cofactor expansion (choose a convenient row/column).

3.2 Without fully expanding, compute $\det(A)$ for

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 5 \end{bmatrix}.$$

Explain why your method works.

3.3 Use determinant properties to compute $\det(B)$ given $\det(A) = 6$:

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix} A.$$

3.4 Use Cramer's rule to solve

$$\begin{cases} 2x + y = 5, \\ 3x - y = 4. \end{cases}$$

3.5 (Determinant and invertibility) Consider

$$C = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{bmatrix}.$$

Compute $\det(C)$ quickly (hint: look for row dependence). Is C invertible?

Part IV: Vector Spaces (Chapter 4: 4.1–4.2)

1. Vector Spaces and Subspaces (4.1)

4.1 Is the set

$$S = \{(x, y) \in \mathbb{R}^2 : x + y = 1\}$$

a subspace of $\mathbb{R}^2$? If not, identify which subspace property fails.

4.2 Let

$$W = \{(x, y, z) \in \mathbb{R}^3 : x - 2y + z = 0\}.$$

- (a) Prove W is a subspace of $\mathbb{R}^3$ using the subspace test.
 - (b) Find a parametric vector description of W .
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2. Null Space and Column Space (4.2)**4.3** For

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix},$$

find $\text{Null}(A)$ and $\text{Col}(A)$, and compute $\text{rank}(A)$.

4.4 Let

$$M = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

- (a) Find a basis for $\text{Col}(M)$.
 - (b) Find a basis for $\text{Null}(M)$.
 - (c) Verify $\dim(\text{Col}(M)) + \dim(\text{Null}(M)) = 3$.
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Appendix: Solutions

Solutions to Part I

1.1 Solve:

$$\begin{cases} x + 2y - z = 1 \\ 2x + 5y - 3z = 4 \\ x + 3y - 2z = 2 \end{cases}$$

Subtract (1) from (3): $y - z = 1$. Subtract $2 \cdot (1)$ from (2): $y - z = 2$. Contradiction, so the system is **inconsistent**. No solutions.

1.2 Second equation has same left-hand side as $2 \cdot (\text{first})$, but RHS differs: $2 \cdot 0 = 0 \neq 3$. Inconsistent. No solutions.

1.3 Row reduce:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 2 & 4 & 7 & 3 \\ 1 & 2 & 4 & 2 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

Thus $z = 1$ and $x + 2y = -1$. Let $y = t$.

$$x = -1 - 2t, \quad y = t, \quad z = 1.$$

Solution set:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}.$$

Pivot columns: 1 and 3. Free variable: y .

1.4 Consistency requires $k = 2$ (since row 2 is $2 \cdot$ row 1 on the left). If $k = 2$, infinitely many solutions:

$$x + y + z = 1 \Rightarrow x = 1 - y - z.$$

Let $y = s, z = t$:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

If $k \neq 2$, no solutions.

1.5 Solve $c_1 v_1 + c_2 v_2 = b$:

$$c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} \Rightarrow \begin{cases} c_1 = 1 \\ c_1 + c_2 = 3 \\ c_2 = 2 \end{cases}$$

Works. So $b = v_1 + 2v_2$.

1.6 (a)

$$Ax = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 + 2x_3 \\ 2x_1 + x_2 + 3x_3 \end{bmatrix}.$$

Also $Ax = x_1 a_1 + x_2 a_2 + x_3 a_3$ where a_i are columns. (b) Since the columns span $\mathbb{R}^2$ (e.g. columns 1 and 2 are independent), $Ax = b$ is consistent for all $b \in \mathbb{R}^2$.

1.7 Equations are identical; one independent equation: $x + 2y - z = 0$. Let $y = s, z = t$:

$$x = -2s + t, \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

1.8 Solve $Ax = b$ with

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}.$$

Equations: $x_1 + 2x_2 = 3$, $x_2 + x_3 = 2$. Let $x_3 = t$. Then $x_2 = 2 - t$, and $x_1 = 3 - 2(2 - t) = -1 + 2t$. A particular solution: take $t = 0$:

$$x_p = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}.$$

Homogeneous: $x_1 + 2x_2 = 0$, $x_2 + x_3 = 0$. Let $x_3 = t$, then $x_2 = -t$, $x_1 = 2t$:

$$x_h = t \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}.$$

So $x = x_p + x_h$.

1.9 Dependent, since the second vector is 2 times the first:

$$\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} - 2 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \mathbf{0}.$$

1.10 Standard matrix from coefficients:

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}.$$

Then $T(1, 2) = (2 \cdot 1 - 2, 1 + 6) = (0, 7)$. $\det(A) = 2 \cdot 3 - (-1) \cdot 1 = 7 \neq 0$, so T is one-to-one and onto.

1.11

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}, \quad T(2, -1) = (2 + 2(-1), 6 - (-1)) = (0, 7).$$

Solutions to Part II

2.1

$$\begin{aligned} A + B &= \begin{bmatrix} 3 & 2 \\ 4 & 5 \end{bmatrix}, & 3A &= \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix}, \\ AB &= \begin{bmatrix} 1 \cdot 2 + 2 \cdot 1 & 1 \cdot 0 + 2 \cdot 1 \\ 3 \cdot 2 + 4 \cdot 1 & 3 \cdot 0 + 4 \cdot 1 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 10 & 4 \end{bmatrix}, \\ BA &= \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 4 & 6 \end{bmatrix}. \end{aligned}$$

Not equal.

2.2

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \Rightarrow A^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I, \quad A^3 = A^2 A = -A.$$

Pattern: $A^{2n} = (-1)^n I$ and $A^{2n+1} = (-1)^n A$.

2.3 For $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$,

$$\det(A) = 2 \cdot 3 - 5 \cdot 1 = 1, \quad A^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}.$$

Then

$$x = A^{-1}b = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}.$$

2.4 (a) True (for square A). (b) True. (c) True.

Solutions to Part III

3.1 Expand along the first row:

$$\det = 1 \det \begin{bmatrix} 4 & 5 \\ 1 & 2 \end{bmatrix} - 2 \det \begin{bmatrix} 3 & 5 \\ 0 & 2 \end{bmatrix} = 1(8 - 5) - 2(6 - 0) = 3 - 12 = -9.$$

3.2 Upper triangular, so determinant is product of diagonal:

$$\det(A) = 2 \cdot 3 \cdot 5 = 30.$$

3.3 $\det(B) = \det(\text{diag}(1, 2, -1)) \det(A) = (1 \cdot 2 \cdot -1) \cdot 6 = -12$.

3.4 Cramer's rule:

$$\Delta = \det \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix} = -2 - 3 = -5.$$

$$\Delta_x = \det \begin{bmatrix} 5 & 1 \\ 4 & -1 \end{bmatrix} = -5 - 4 = -9 \Rightarrow x = \frac{\Delta_x}{\Delta} = \frac{-9}{-5} = \frac{9}{5}.$$

$$\Delta_y = \det \begin{bmatrix} 2 & 5 \\ 3 & 4 \end{bmatrix} = 8 - 15 = -7 \Rightarrow y = \frac{\Delta_y}{\Delta} = \frac{-7}{-5} = \frac{7}{5}.$$

3.5 Rows 1 and 2 are dependent (row 2 = 2 · row 1), so $\det(C) = 0$. Thus C is not invertible.

Solutions to Part IV

4.1 Not a subspace: it does not contain the zero vector since $(0, 0)$ does not satisfy $x + y = 1$.

4.2 (a) Let $u, v \in W$ so $u_1 - 2u_2 + u_3 = 0$ and $v_1 - 2v_2 + v_3 = 0$. Then $(u+v)_1 - 2(u+v)_2 + (u+v)_3 = 0$, so closed under addition. For scalar c , $(cu)_1 - 2(cu)_2 + (cu)_3 = c(u_1 - 2u_2 + u_3) = 0$, so closed under scalar mult. Hence W is a subspace.

(b) Solve $x - 2y + z = 0 \Rightarrow x = 2y - z$. Let $y = s, z = t$:

$$(x, y, z) = (2s - t, s, t) = s(2, 1, 0) + t(-1, 0, 1).$$

4.3 For $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$, row 2 is 2 · row 1, so rank is 1. Null space solves $x_1 + 2x_2 + 3x_3 = 0$. Let $x_2 = s, x_3 = t$, then $x_1 = -2s - 3t$:

$$\text{Null}(A) = \text{span} \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Column space is spanned by the first column (others are multiples):

$$\text{Col}(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}.$$

Thus $\text{rank}(A) = 1$.

4.4 For

$$M = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix},$$

pivot columns are 1 and 2, so a basis for $\text{Col}(M)$ is

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

Null space: solve $x_1 + 2x_2 + x_3 = 0$, $x_2 + x_3 = 0$. Let $x_3 = t$, then $x_2 = -t$, $x_1 = t$:

$$\text{Null}(M) = \text{span} \left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

Dimensions: $\dim(\text{Col}(M)) = 2$, $\dim(\text{Null}(M)) = 1$, sum is 3.